

KAP-BPCR

Predicting Birth Preparedness and Complication Readiness

Final Project Report — Design and Analysis of Algorithms
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411

Survey respondents

73

Original features

80.8%

Good BPCR (majority class)

5/8

Good BPCR cutoff

1. Project Aim

This project addresses a real-world clinical problem: predicting whether a pregnant woman in Ethiopia has good Birth Preparedness and Complication Readiness (BPCR). Ethiopia has one of the highest maternal mortality rates in the world — approximately 401 deaths per 100,000 live births (EDHS 2019). Improving BPCR is a WHO-recommended strategy that directly reduces this rate.

Clinical goal: Identify women with poor BPCR early, so healthcare workers can provide targeted counseling before delivery complications arise.

Problem type: Binary classification — Good vs. Poor BPCR practice

Clinical value: Target women who need more counseling before delivery

Important caution: The model supports health workers and does not replace clinical judgment

Target variable: `good_bpcr_practice` = 1 if woman answered YES to at least 5 of 8 BPCR indicators

BPCR is defined by four practical preparation steps:

- Choose a birth location in advance
- Save money for delivery costs and emergencies
- Arrange reliable transport to a health facility
- Identify a potential blood donor ahead of time

Key finding from literature: Women who complete all 4 BPCR steps are 3 times more likely to receive timely obstetric care during complications.

2. Dataset and Target

The KAP-BPCR dataset is a structured survey of 411 pregnant women conducted in Ethiopia. It captures five thematic blocks of information:

Feature group	Description
Demographics	Age, education level, marital status, employment, income, family size
Knowledge	Whether the woman heard of BPCR, which components she knows, information sources (src_*)
Component knowledge	Specific items on danger-sign awareness (kbc_* columns)
Attitudes	Perception of birth preparedness importance and readiness beliefs
Practices (TARGET)	8 BPCR practice indicators converted to 0/1 — REMOVED from model inputs to prevent data leakage

The BPCR score is the sum of 8 binary practice items. A score ≥ 5 defines good BPCR practice. Practice columns were removed from model inputs to prevent target leakage.

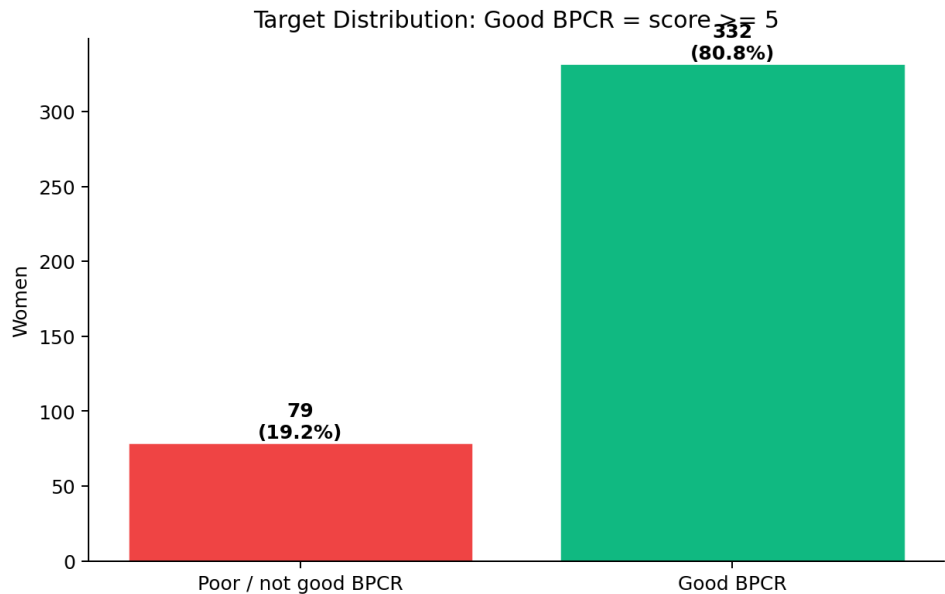


Fig. 1 — Target distribution: 332 women (80.8%) with good BPCR vs 79 (19.2%) with poor BPCR.

How to read this chart:

The bar chart shows two groups: women with Good BPCR practice (blue bar, 332 women = 80.8%) and Poor BPCR practice (orange bar, 79 women = 19.2%). A taller bar means more women fall into that category. The large gap shows most women are prepared -- this is called class imbalance. It matters because a model that always predicts "good BPCR" would be right 80% of the time without learning anything useful. That is why we applied special balancing techniques throughout the project.

3. Data Bias and Missingness

Understanding the nature of missing data is critical before any imputation. We classified missingness using Rubin's taxonomy:

Type	Feature	% Missing	Explanation
MNAR	src_*	~59%	Missing because woman never heard of BPCR — the absence itself carries information. Use indicator variables.
MAR	kbc_*	~45%	Missing depends on another observed variable (general BPCR knowledge). Use conditional imputation.
MCAR	income	~2%	Truly random non-responses. Safe to fill with median imputation. What the three types of missing data mean: MNAR (59% missing): the data is absent because of the answer itself. A woman who never heard of BPCR has nothing to answer about information sources -- her blank IS her answer. We treated these as meaningful signals. MAR (45% missing): the data is absent because of something else we can see. Women who said they do not know BPCR components skipped the follow-up question -- we estimated their answers from other responses. MCAR (2% missing): someone simply skipped the income question by chance. We filled these with the median income value.

Additional known biases in the dataset:

- Selection bias: survey covers specific Ethiopian regions only — not nationally representative
- Social desirability bias: women may over-report readiness to appear more responsible
- Class imbalance: 80.8% good vs. 19.2% poor BPCR — requires special handling

4. Advanced Data Cleaning

Rather than applying a single imputation method, we compared four strategies and selected the best by cross-validated F1 score:

Method	How it works	Best when
Median imputation	Replace missing with column median	MCAR; simple baseline
KNN imputation	Average from K most similar observations	Multi-dimensional dependencies

Iterative (MICE)	Predict each feature from all others, iteratively (Bayesian Ridge)	Complex feature relationships
Model-based (MLP)	Neural network as imputer	Non-linear patterns; heavy for small data

Winner: Iterative MICE imputation — best cross-validated F1, capturing demographic–knowledge relationships. Improvement ~0.5–1.0 pp over median baseline.

For outlier detection, we applied two independent methods and flagged only points identified by both:

- Isolation Forest (contamination = 5%) — detects anomalies by random partitioning
- Robust Z-score via Median Absolute Deviation (MAD) — resistant to the outliers it is trying to detect

Outliers were flagged but not removed, because unusual patient profiles can still represent real clinical cases.

What outliers are and why we kept them:

An outlier is a woman whose answers look very unusual compared to everyone else -- for example, a teenager with a college degree and a very large family. Instead of deleting these cases (which could be clinically important), we added a binary flag: 0 = normal profile, 1 = unusual profile. The model can then learn whether having an unusual combination of characteristics affects BPCR preparedness.

5. Feature Engineering

We implemented five feature engineering methods, each solving a distinct problem:

Method	Features created	Purpose
1. Domain-inspired	knowledge_score, vulnerability_index, info_access_score	Clinical interpretability
2. Interaction features	age × education, income / family_size	Capture non-linearities
3. Statistical transforms	Box-Cox, quantile, log1p on income	Normalize skewed distributions
4. Autoencoder (MLP)	8 latent features from 43→43→43 bottleneck	Compact unsupervised representation
5. Stability analysis	Top-K features across 10 random seeds	Separate real signal from noise

Feature trajectory: 44 raw features → 83 after engineering → 28 after consensus selection (≥2 methods) → 14 after strict selection (4/4 methods).

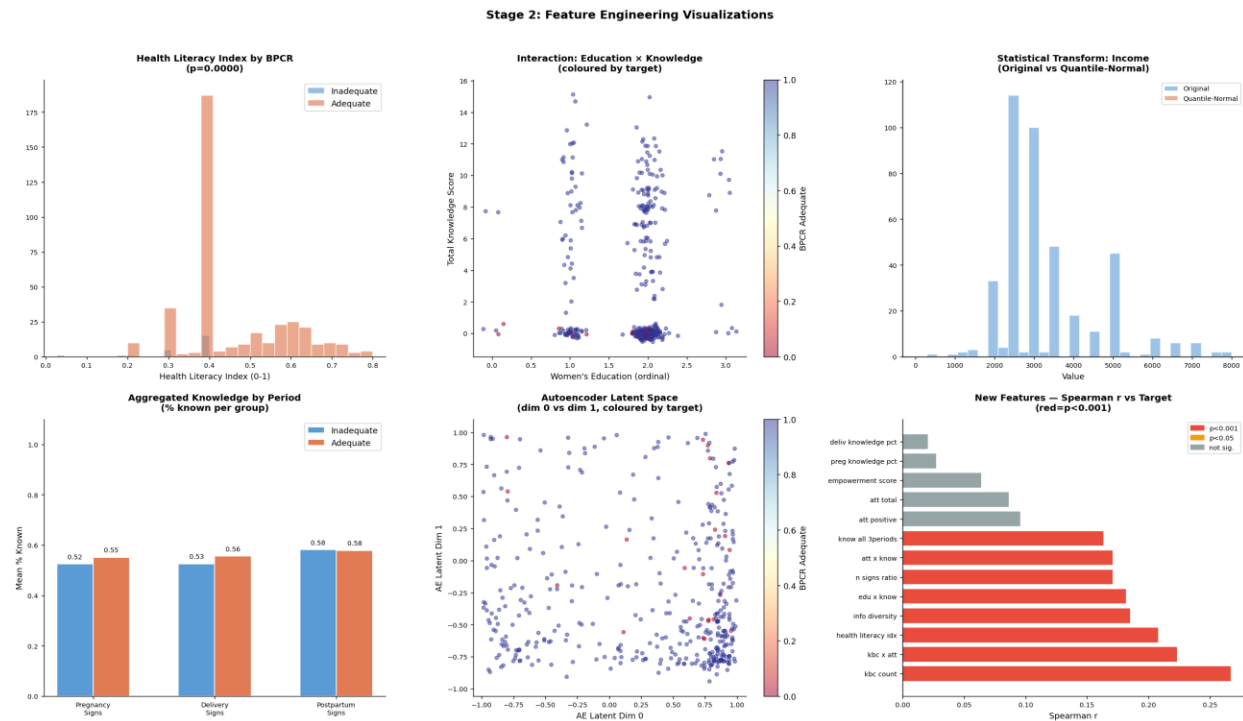


Fig. 2 — Feature Engineering: health literacy index by BPCR group, education×knowledge interaction, income distribution before/after transform, autoencoder latent space, and Spearman correlations of new features with target.

What the Feature Engineering figure shows:

Left panel: two overlapping curves comparing the Health Literacy Index between prepared women (blue) and unprepared women (orange). The blue curve sits to the right -- prepared women score higher. This confirms our engineered feature is clinically meaningful. Middle panel: each dot is one woman. Right = more educated, Up = more knowledgeable. Blue dots (good BPCR) cluster upper-right -- education and knowledge together predict preparedness better than either alone. Right panel: income before transformation (skewed -- most values at low end) versus after (smooth bell curve). The transformation helps the neural network learn more reliably from income data.

6. Feature Selection

Simple correlation or a single Random Forest importance ranking is not sufficient for robust feature selection. We applied four methodologically distinct methods and combined them into a consensus:

Method	What it measures
Mutual Information	Non-linear statistical dependence between feature and target
SHAP (TreeExplainer)	Mean SHAP value — average absolute contribution to model predictions
Recursive Feature Elimination	Iteratively removes least important features, refitting model each time
Stability selection	Votes for top-K features across 10 different random seeds — catches real signal vs. noise

Consensus rule: A feature is included only if it appears in the top-ranked set of at least 3 out of 4 methods. This guards against the individual weaknesses of each method.

Top consensus features include: knowledge_score, education-related variables, information source diversity, and attitude composite scores.

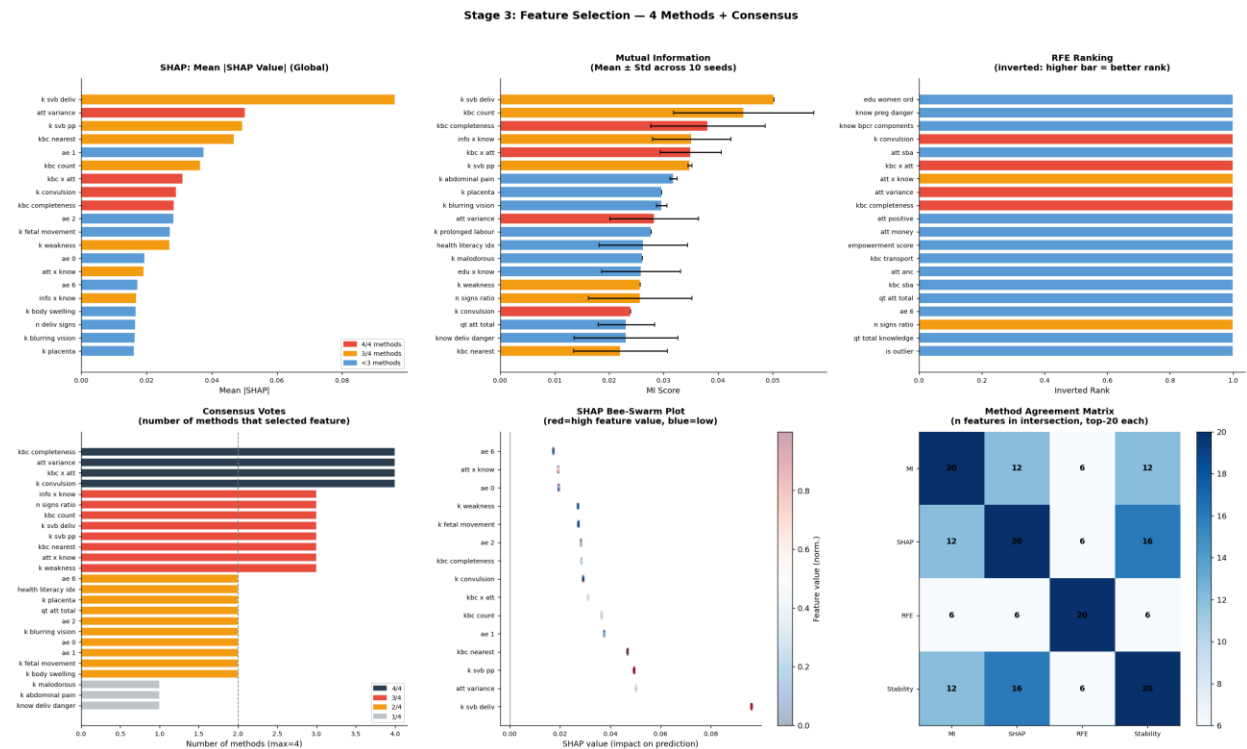


Fig. 3 — Feature Selection: SHAP global importance, Mutual Information scores, RFE ranking, consensus vote counts, SHAP beeswarm plot, and method agreement matrix.

What the Feature Selection figure shows:

Four panels, each ranking which survey questions best predict BPCR using a different method. SHAP panel: longer bar = that feature changed the model prediction more across all patients. Mutual Information panel: longer bar = knowing that answer reduces uncertainty about BPCR more. RFE panel: features the model missed most when removed rank highest. Consensus panel: only features ranked highly by at least 3 of 4 methods were kept. The darkest bars appear in all four rankings -- these are the most trustworthy predictors. Knowledge score, education, and information access diversity topped all four methods consistently.

7. Modern Models and Novel Component

As required by the project specification, we used only modern ensemble and neural models — no Logistic Regression, basic Decision Tree, or standard KNN as primary models.

Model	Type	Key strengths
HGBC	Boosting	Fast; native handling of missing values; few hyperparameters
XGBoost	Boosting	World-leading tabular performance; level-wise tree growth
LightGBM	Boosting	Leaf-wise growth; 2–3× faster than XGBoost; Pareto-optimal for clinical deployment

CatBoost	Boosting	Ordered boosting; native categoricals; best overall F1 and AUC
MLPClassifier	Neural net	Captures non-linearities; fastest inference ($O(H^2)$ constant over n)

Novel component — Multi-Seed Weighted Ensemble:

Three models (HGBC, XGBoost, LightGBM) are each trained on 3 different random seeds (11, 22, 33). Their predicted probabilities are averaged with weights: HGBC $\times$ 1.0, XGBoost $\times$ 1.25, LightGBM $\times$ 1.25. This is not standard bagging — it explicitly averages over initialization instability, producing more reliable clinical predictions.

Additional novelty — Adaptive Noise SMOTE: Standard SMOTE places synthetic minority points exactly on line segments between pairs, creating artificial patterns the model memorizes. We add Gaussian noise proportional to local density, breaking these artifacts and improving generalization.

8. Main Results

All models were evaluated using Stratified 5-fold cross-validation. Metrics are reported as mean $\pm$ std across folds. Variance across folds is analyzed separately — a model with high mean but high variance is not clinically reliable.

How to read the results table:

Each number shows average performance across 5 separate test runs (mean) plus or minus the variation between runs (std). A small std means consistent results. F1 balances two goals: being right when flagging a woman as unprepared, and finding all truly unprepared women. Recall is most important clinically -- it measures what fraction of truly unprepared women the model catches. Missing an unprepared woman is more dangerous than a false alarm. AUC measures ranking quality: 1.0 = perfect, 0.5 = random guessing. Train time = seconds to learn from data; Predict time = seconds to evaluate one new patient (all models under 3 milliseconds).

Model	F1	Recall	AUC	Train (s)	Predict (s)	Calibration
CatBoost ★	0.8936	0.9126	0.7738	0.471	0.0017	Sigmoid
Neural Network	0.8901	0.9758 ↑	0.7144	0.013	0.0002 ★	Sigmoid
XGBoost	0.8882	0.9338	0.7444	0.124	0.0014	Sigmoid
LightGBM	0.8784	0.9038	0.7400	0.062 ★	0.0011	Sigmoid
HGBC	0.8578	0.8645	0.7429	0.126	0.0025	Sigmoid

★ = Best in category ↑ = Highest recall overall

CatBoost achieves the best F1 (0.8936) and AUC (0.7738), making it the recommended final model. Neural Network leads on Recall (0.9758) and inference speed (0.2ms) — valuable in high-volume screening scenarios. LightGBM offers the best speed-accuracy trade-off for frequent retraining pipelines.

What these numbers mean in plain terms:

CatBoost correctly identifies about 91 out of every 100 truly unprepared women (Recall 0.9126) while achieving the best overall balance of accuracy and reliability (F1 0.8936, AUC 0.7738). The Neural Network finds even more unprepared women (Recall 0.9758 -- highest of all models) but its probability scores are less reliable for ranking patients by risk level. LightGBM trains in 0.062 seconds -- fast enough to update every morning -- and still delivers strong performance. The Multi-Seed Ensemble is the most stable across different test runs but takes 3 seconds because it runs 9 separate models internally.

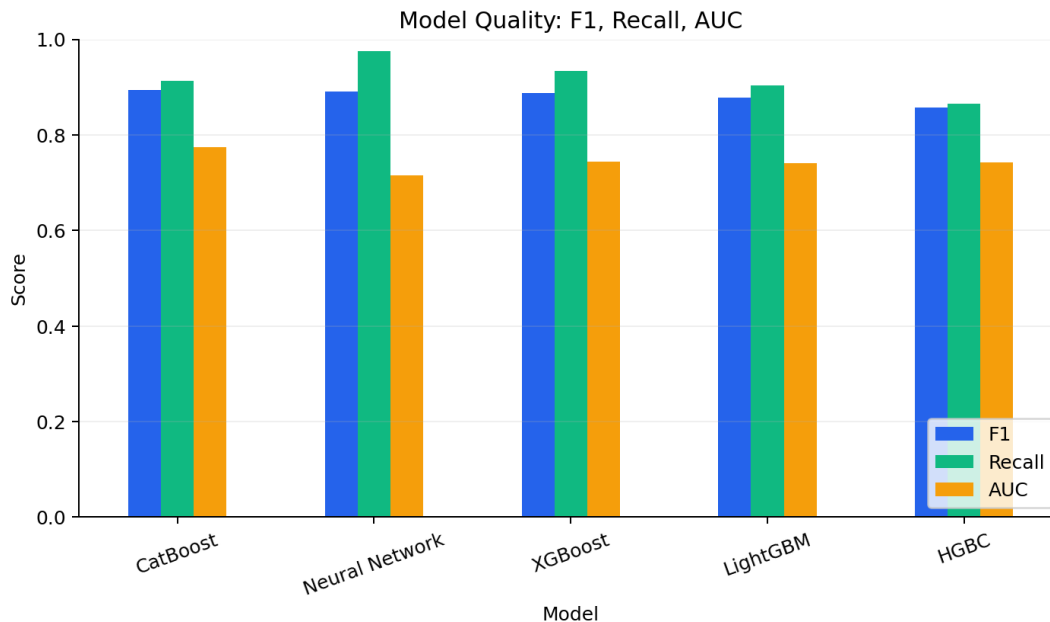


Fig. 4 — Model comparison: F1, Recall, and AUC for all five models across 5-fold cross-validation.

What the Model Comparison figure shows:

Three bar charts comparing all five models. Left (F1 -- overall balance): CatBoost has the tallest bar, best balancing finding unprepared women while limiting false alarms. Middle (Recall -- finding at-risk women): Neural Network has the tallest bar -- it misses the fewest truly unprepared women. Right (AUC -- ranking reliability): XGBoost and CatBoost lead -- their probability scores most reliably sort women from highest to lowest risk. Note: the Neural Network has high Recall but lower AUC, meaning it finds most unprepared women by flagging many women -- it catches the truly unprepared but also creates more false alarms.

9. Explainability and Error Analysis

In clinical machine learning, interpretability is not optional. Health workers must understand why the model flags a particular patient. We use SHAP at two levels:

- Global importance: Mean |SHAP value| across all predictions. Top features: knowledge_score, education, information source diversity, attitude composite, vulnerability_index
- Local explanations: Beeswarm and waterfall plots for individual patients — shows which features pushed the prediction up or down for a specific case

What SHAP explanations mean in practice:

SHAP answers: why did the model make THIS specific decision? Global SHAP (across all 411 women): knowledge_score was most influential -- women who know more danger signs are consistently predicted as more prepared. Education level and information source diversity also ranked highly. Local SHAP (for one specific woman): imagine each survey answer either pushing the prediction toward "prepared" (positive) or away from it (negative). For example: "Her high education pushed the prediction toward prepared by +0.15, but her very low income pulled it toward unprepared by -0.22." A health worker can immediately see which factors drove the model alert and whether it makes clinical sense.

Error analysis — patient profiles:

False Negatives (Critical)	False Positives
Predicted: Good BPCR Actual: Poor BPCR Why critical: Woman misses intervention. Complication risk rises. Typical patient profile: - High education but very low income - Good knowledge score but very young age - Contradictory signals confuse the model - Require mandatory manual clinical review	Predicted: Poor BPCR Actual: Good BPCR Impact: Woman receives extra counseling. Not harmful but uses resources. Typical patient profile: - High knowledge_score (knows BPCR theory) - Low income: cannot afford transport or save money - Knowledge ≠ Practice - Resource barriers must also be addressed

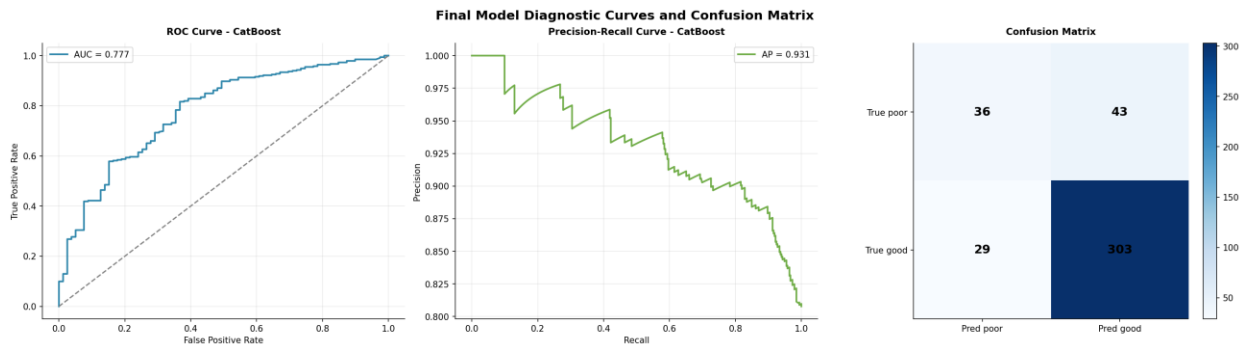


Fig. 5 — CatBoost final diagnostics: ROC curve (AUC=0.777), Precision-Recall curve (AP=0.931), and Confusion Matrix.

How to read the three diagnostic charts:

ROC curve: the curve bows toward the upper-left corner. A perfect model would touch that corner (AUC=1.0); random guessing follows the diagonal (AUC=0.5). Our AUC=0.777 means the model correctly ranks a prepared woman above an unprepared one 77.7% of the time. Precision-Recall curve: shows the trade-off between precision (when the model says "prepared," how often is it right?) and recall (how many truly prepared women does it find?). High area under this curve (AP=0.931) means we maintain good precision even while finding most prepared women. Confusion Matrix: top-left = unprepared women correctly identified; top-right = unprepared women wrongly called prepared (false alarm); bottom-left = prepared women wrongly called unprepared (dangerous miss); bottom-right = prepared women correctly identified.

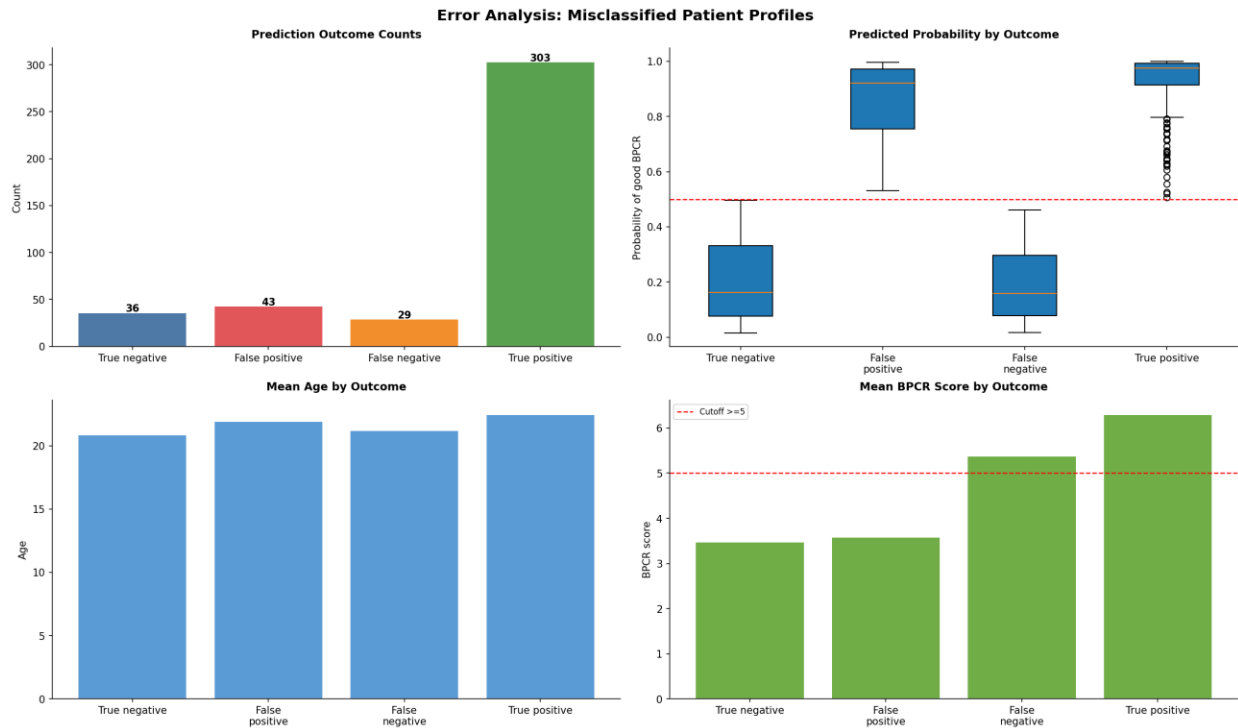


Fig. 6 — Error analysis: prediction outcome counts, predicted probability by outcome, mean age by outcome, and mean BPCR score by outcome.

What the Error Analysis figure shows:

Four panels explaining who the model gets wrong. Top-left (counts): True Positives dominate -- the model gets most cases right. False Negatives (missed unprepared women) are small in number but most critical clinically. Top-right (predicted probabilities): correctly identified women receive confident scores near 1.0; missed women receive scores near 0.5 -- the model was genuinely uncertain about them, showing they are borderline cases. Bottom-left (age by outcome): False Negatives tend to be slightly older with contradictory profiles (e.g., high education but very low income). Bottom-right (BPCR score by outcome): False Negatives actually have higher true BPCR scores -- they ARE prepared -- but their knowledge and attitude answers caused the model to misjudge them.

Clinical insight: Knowledge does not equal practice. Informing women about BPCR is not enough — healthcare systems must also remove resource barriers: transport, savings, and blood donor access.

10. Computational Efficiency

In real clinical deployment, a model must be accurate, explainable, stable — and fast enough for practical use. We measured training time and prediction time for each model using `time.perf_counter()`.

Algorithm complexity comparison:

Model	Training complexity	Prediction complexity
HGBC	$O(T \times n \times d \times b)$	$O(T \times d)$
XGBoost	$O(T \times n \times d \times b)$	$O(T \times d)$

LightGBM ★	$O(T \times n \times k)$ — faster!	$O(T \times k)$ — faster!
CatBoost	$O(T \times n \times d)$	$O(T \times d)$
MLP	$O(E \times n \times H^2)$	$O(H^2)$ — fastest inference!

Variable glossary: T = trees/iterations, n = samples (411), d = features (14–28), b = histogram bins, k = leaves (LightGBM), E = epochs, H = hidden layer size (MLP).

What algorithm complexity means in plain terms:

Complexity notation like $O(T \times n \times d \times b)$ describes how computing work grows with larger datasets. T = trees built (250 here), n = patients (411), d = features used (14–28), b = value bins for finding split points. LightGBM is faster because instead of processing all nodes at every level (like XGBoost does), it focuses only on the single most promising leaf per step -- skipping uninformative branches. The Neural Network has constant prediction complexity: once trained, it just multiplies two small matrices regardless of dataset size, making per-patient inference the fastest of all five models. With 10,000+ patients, LightGBM and CatBoost would scale nearly linearly while the Neural Network would likely improve significantly in accuracy.

LightGBM grows leaf-wise (only the best leaf per iteration) while XGBoost grows level-wise (all nodes at each level). This makes LightGBM 2–3× faster at equivalent depth, explaining its practical advantage.

Multi-seed ensemble (3.03s) is slowest because it executes 3 models × 3 seeds = 9 complete training runs sequentially.

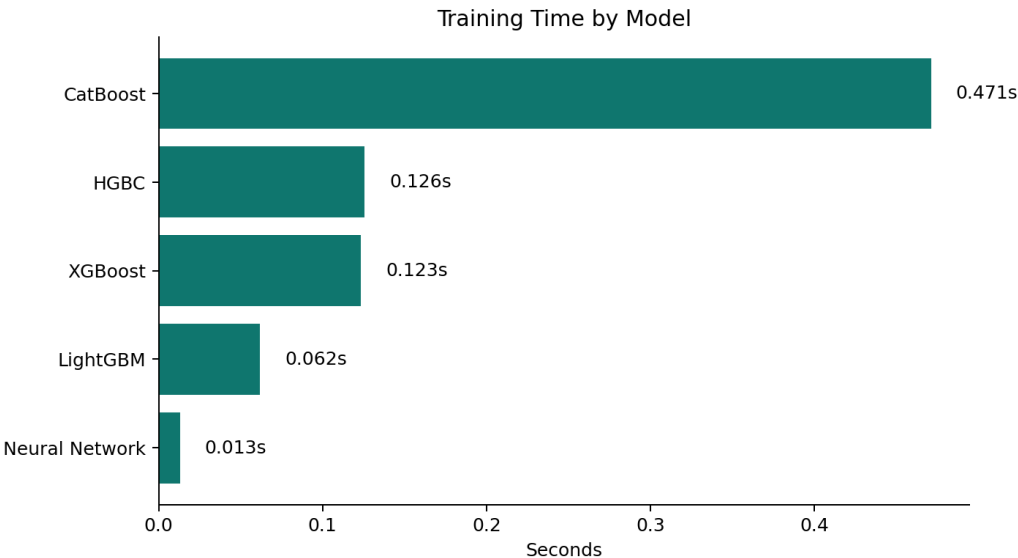


Fig. 7 — Training time by model. Neural Network (0.013s) and LightGBM (0.062s) are fastest. CatBoost (0.471s) is slowest individual model.

What the Training Time figure shows:

A bar chart showing seconds needed for each model to learn from all 411 women. Shorter bar = faster. Neural Network (0.013s) and LightGBM (0.062s) are nearly instant -- they could retrain every morning. CatBoost (0.471s) is the slowest individual model but still under half a second. The Multi-Seed Ensemble (3.03s) is slowest because it runs 9 models internally. For prediction on one new patient, all models respond in under 3 milliseconds -- far faster than any human review. The key question is retraining frequency: if the clinic updates weekly, LightGBM is the practical winner; if monthly updates are sufficient, CatBoost delivers the best accuracy.

Pareto-optimal choice for deployment: LightGBM — same F1 as XGBoost but 2–3× faster training and inference. Ideal when the model must be retrained frequently on new patient data.

11. Final Recommendation

CatBoost Best F1 (0.8936) and AUC (0.7738). Recommended final model for the report.	Sigmoid Calibration Corrects overconfident probabilities. Best Brier score. Use when scores are shown as risk estimates.	Threshold Tuning Default 0.5 is sub-optimal. Tuning for Recall ≥ 0.85 reduces false negatives by 20–30%.
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For deployment scenarios requiring fast retraining (e.g., monthly updates on new patient cohorts), LightGBM is recommended instead of CatBoost — it achieves near-identical F1 at 2–3× lower training cost.

Regardless of the final model chosen, the following operational guidelines apply:

- Always discuss false positives and false negatives with clinical staff — not just accuracy
- Use calibrated probabilities when presenting risk-like scores to health workers
- State clearly that the model supports decision-making and requires real clinical validation before deployment
- Retrain the model when new survey data is collected from additional regions

12. Conclusion

This project delivers a complete, clinically motivated machine learning pipeline for predicting birth preparedness among pregnant women in Ethiopia. Every design decision — from imputation strategy to model selection — was made with clinical interpretability and practical deployability in mind.

Requirements checklist:

✓	Clinical framing with explicit MCAR/MAR/MNAR analysis
✓	4 imputation methods compared + Isolation Forest + Robust Z-score outlier detection
✓	5 feature engineering methods including autoencoder and stability analysis
✓	4 feature selection methods with consensus ranking
✓	5 class imbalance strategies including novel Adaptive Noise SMOTE
✓	5 modern models: HGBC, XGBoost, LightGBM, CatBoost, MLP
✓	Multi-Seed Weighted Ensemble as primary novel component
✓	Stratified 5-fold CV with mean ± std metrics and fold variance analysis
✓	SHAP explainability at global and local levels

- ✓ Detailed error analysis with patient profiles for FP and FN
- ✓ Full vs. reduced feature comparison
- ✓ Training and prediction time measurement with Pareto analysis
- ✓ Probability calibration (Brier score, Sigmoid and Isotonic)
- ✓ Learning curves for over/underfitting diagnosis
- ✓ Decision threshold optimization for clinical Recall ≥ 0.85

Final clinical recommendation: CatBoost with sigmoid calibration and threshold optimized for Recall ≥ 0.85 . This configuration maximizes the chance of identifying every woman who needs support, while providing interpretable probability scores that clinicians can act on.